Probability distributions and the Gaussian

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Every method in this course either estimates a probability distribution or makes predictions that are best understood through one. This note collects the probability we need — how distributions are described, the four things we do with them, and the multivariate Gaussian in detail — and connects it to two pillars of modern machine learning: maximum-likelihood training of language models and Gaussian noise in diffusion models. Notation follows Note 1, Section 2.

1 Describing a distribution

A random variable X is described by how its probability is spread over its possible values. Three descriptions are in common use.

Cumulative distribution function. For a real-valued X, the c.d.f. is $F(x) = P(X \le x)$. It is non-decreasing, tends to 0 as $x \to -\infty$ and to 1 as $x \to +\infty$, and it exists for every random variable, discrete or continuous. For a vector $\mathbf{X} = (X_1, \dots, X_d)$ the inequality is read component-wise: $F(\mathbf{x}) = P(X_1 \le x_1, \dots, X_d \le x_d)$.

Probability table (discrete variables). If X takes finitely or countably many values, its distribution is the list of probabilities $P(X = x)$, which are non-negative and sum to one. For several discrete variables the table becomes an array with one axis per variable; a joint table for d variables with m values each has m^d entries — a first sign of the difficulty Note 3 studies.

Density (continuous variables). If F is differentiable, the probability density function is $p(x) = F'(x)$, and probabilities of regions are integrals,

$$P(\mathbf{X} \in A) = \int_A p(\mathbf{x})\, d\mathbf{x}, \qquad \int_{\mathbb{R}^d} p(\mathbf{x})\, d\mathbf{x} = 1.$$

A density is not a probability: it carries units of $1/[x]$ and may exceed one. A Gaussian of width $\sigma = 0.1$ mm has a peak density of about 4 mm^{-1}.

The empirical distribution. Given a sample $x_1, \dots, x_n$, the empirical c.d.f. $\hat{F}(x) = \frac{1}{n}\#\{i : x_i \le x\}$ is a step function that jumps by $1/n$ at each data point. By the law of large numbers $\hat{F}(x) \to F(x)$ for every x; the convergence is even uniform in x (Glivenko–Cantelli). The empirical distribution is the simplest estimate of a distribution, and every learning algorithm can be viewed as a way of smoothing it.

1.1 Joint, marginal and conditional

For two variables with joint density $p(\mathbf{x}, \mathbf{z})$ (sums replace integrals in the discrete case):

$$\begin{aligned} \text{marginal:} \quad & p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z})\, d\mathbf{z}, \\ \text{conditional:} \quad & p(\mathbf{z} \mid \mathbf{x}) = \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{x})} \qquad (p(\mathbf{x}) > 0), \\ \text{Bayes' rule:} \quad & p(\mathbf{z} \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})}{\int p(\mathbf{x} \mid \mathbf{z}')\, p(\mathbf{z}')\, d\mathbf{z}'}. \end{aligned}$$

Independence means $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x})\, p(\mathbf{z})$, equivalently $p(\mathbf{z} \mid \mathbf{x}) = p(\mathbf{z})$. Expectations are integrals against the density, $\mathbb{E}[g(\mathbf{X})] = \int g(\mathbf{x})\, p(\mathbf{x})\, d\mathbf{x}$; the mean is $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ and the covariance matrix is $\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\mathsf{T}}]$, with variances on its diagonal.

2 Four things we do with a distribution

Whether the distribution is a probability table, a Gaussian or a neural network with billions of parameters, the same four operations recur.

1. **Sample**: generate new data $\mathbf{x} \sim p$. This is what a generative model is for.
2. **Evaluate**: compute $p(\mathbf{x})$ (the *likelihood* of a configuration), possibly after marginalising over unobserved variables. Low likelihood flags unusual data.
3. **Infer**: given some variables, predict others — the most probable value or the conditional mean $\mathbb{E}[\mathbf{Z} \mid \mathbf{X} = \mathbf{x}]$. Supervised learning is inference of Y from $\mathbf{X}$.
4. **Learn**: choose the parameters θ of a family p_θ from data. The standard principle is *maximum likelihood*,

$$\hat{\theta} = \arg\max_{\theta} \prod_{i=1}^{n} p_\theta(\mathbf{x}_i) = \arg\min_{\theta} \; -\frac{1}{n} \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i), \tag{1}$$

   the product being justified by the i.i.d. assumption. The right-hand side is the average negative log-likelihood, the density-estimation loss of Note 1, Table 2.

Example 1 (a categorical variable). The number $X \in \{0, 1, 2\}$ of reconstructed jets in a simplified event selection is described by three numbers π_0, π_1, π_2 with $\sum_k \pi_k = 1$. *Sampling* draws u uniform on $[0, 1)$ and returns the first k with $\pi_0 + \cdots + \pi_k > u$. *Learning* by maximum likelihood from n events with counts n_0, n_1, n_2 gives $\hat{\pi}_k = n_k / n$, the observed frequencies (Exercise E1).

> **Modern connection — a language model is a categorical distribution**
>
> A large language model defines, for every prefix of text, a categorical distribution over a vocabulary of $V \approx 10^5$ tokens. The network outputs scores ("logits") $z_1, \ldots, z_V$ and the probabilities are $\pi_k = e^{z_k/T} / \sum_j e^{z_j/T}$, where the *temperature* $T = 1$ during training. All four operations appear: training is maximum likelihood (1) on the next token, i.e. minimising cross-entropy; generating text is repeated sampling, with $T < 1$ concentrating mass on likely tokens and $T \to 0$ giving the single most probable token; evaluating the likelihood of a text gives its *perplexity*, exp of the average negative log-likelihood per token; and answering a question is inference, conditioning on the prompt. The example above — learned frequencies from counts — is exactly what such a model would do if it had no parameters to share between contexts.

3 The univariate Gaussian

The Gaussian (normal) density with mean μ and variance σ^2 is

$$\mathcal{N}(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{2}$$

It is a symmetric bump centred at μ with width set by the standard deviation σ (Figure 1). About 68.3%, 95.4% and 99.7% of the probability lies within one, two and three standard deviations of the mean; physicists' "5σ" discovery threshold corresponds to a one-sided tail probability of 2.9×10^{-7}. Any Gaussian is a shifted and scaled standard one: $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0,1)$.

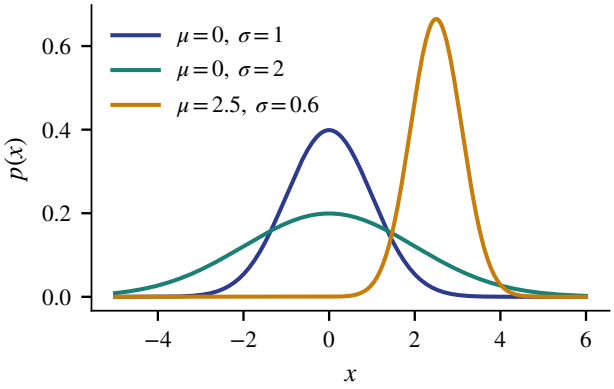


Figure 1. Gaussian densities. Doubling σ halves the peak height, since the area stays one.

Why this particular shape? The central limit theorem says that a sum of many independent contributions with finite variance is approximately Gaussian, whatever the individual distributions. Measurement noise built from many small effects is therefore often close to Gaussian. A second reason is that, among all densities with a given mean and variance, the Gaussian has maximal entropy: it assumes nothing beyond those two numbers.

Gaussian noise and squared error. Suppose targets are generated as $y = f(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0,\sigma^2)$. Then $-\log p(y \mid x) = \frac{(y-f(x))^2}{2\sigma^2} + \frac{1}{2}\log(2\pi\sigma^2)$, and maximising the likelihood of a regression dataset over f is the same as minimising the sum of squared errors. The squared loss of Note 1, Table 2, is maximum likelihood under Gaussian noise; heavier-tailed noise calls for other losses (Exercise E3).

> **In physics — χ^2 fits are Gaussian likelihoods**
>
> Fitting a model f_θ to measurements $y_i \pm \sigma_i$ by minimising $\chi^2 = \sum_i (y_i - f_\theta(x_i))^2/\sigma_i^2$ is maximum likelihood with independent Gaussian errors of known widths. The familiar rule "$\Delta\chi^2 = 1$ gives the 1σ interval" is the statement that $-2\log$ likelihood is quadratic near its minimum [2]. When errors are correlated, the sum becomes the quadratic form of the next section, with the covariance matrix of the measurements.

4 The multivariate Gaussian

For $\mathbf{x} \in \mathbb{R}^d$, the Gaussian with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and symmetric positive-definite covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$ is

$$\mathcal{N}(\mathbf{x};\boldsymbol{\mu},\Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\!\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big), \tag{3}$$

where $|\Sigma|$ is the determinant. The prefactor only normalises the density; all the shape is in the exponent. The quantity

$$\Delta^2(\mathbf{x}) = (\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})$$

is the squared *Mahalanobis distance* of $\mathbf{x}$ from the mean: Euclidean distance after the data have been rescaled so that every direction has unit variance. Surfaces of constant density are the ellipsoids $\Delta^2 = \text{const}$.

Geometry through the eigendecomposition. Write $\Sigma = \sum_{i=1}^{d} \lambda_i \mathbf{v}_i \mathbf{v}_i^{\mathsf{T}}$ with orthonormal eigenvectors $\mathbf{v}_i$ and eigenvalues $\lambda_i > 0$. In the rotated coordinates $u_i = \mathbf{v}_i^{\mathsf{T}}(\mathbf{x}-\boldsymbol{\mu})$ the exponent becomes $\sum_i u_i^2/\lambda_i$, so the density factorises into d independent one-dimensional Gaussians with variances λ_i. The ellipsoid axes point along the eigenvectors with half-lengths proportional to $\sqrt{\lambda_i}$, and $|\Sigma| = \prod_i \lambda_i$ measures the squared volume of the ellipsoid. Three families are common (Figure 2):

Family	Covariance	Free parameters in Σ	Shape
Isotropic	$\sigma^2 I$	1	spheres
Diagonal	$\mathrm{diag}(\sigma_1^2, \dots, \sigma_d^2)$	d	axis-aligned ellipsoids
Full	any positive-definite Σ	$d(d+1)/2$	arbitrary ellipsoids

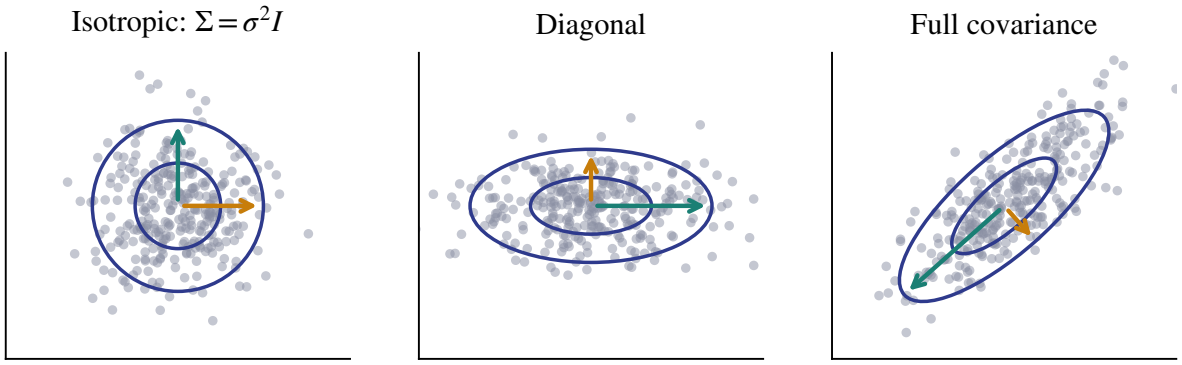


Figure 2. Samples (grey) and the one- and two-σ contour ellipses of three Gaussians in $d = 2$. Arrows are the eigenvectors scaled by $2\sqrt{\lambda_i}$. The isotropic case has no preferred direction; the full covariance captures a correlation between the two coordinates.

The choice is a choice of model size. For $d = 1000$ a full covariance has about half a million parameters, which $n = 500$ examples cannot determine; Note 4 calls this a question of *capacity*.

Closed under marginalising and conditioning. Split $\mathbf{x} = (\mathbf{x}_a, \mathbf{x}_b)$ and partition $\boldsymbol{\mu}$ and Σ accordingly. Every marginal of a Gaussian is Gaussian — just keep the corresponding entries, $\mathbf{x}_a \sim \mathcal{N}(\boldsymbol{\mu}_a, \Sigma_{aa})$ — and every conditional is Gaussian,

$$\mathbf{x}_a \mid \mathbf{x}_b \sim \mathcal{N}\big(\boldsymbol{\mu}_a + \Sigma_{ab}\Sigma_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b),\ \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}\big). \tag{4}$$

The conditional mean is linear in the observed part and the conditional covariance does not depend on it. This is inference in closed form: if two detector channels are correlated, reading one sharpens the prediction for the other by an amount set by the correlation. Formula (4) underlies the Kalman filter used for particle-track reconstruction and spacecraft navigation, and Gaussian-process regression.

5 Sampling: from white noise to anything Gaussian

To sample $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, factor $\Sigma = LL^{\mathsf{T}}$ (Cholesky decomposition, L lower triangular), draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$ — d independent standard normals — and return

$$\mathbf{x} = \boldsymbol{\mu} + L\mathbf{z}. \tag{5}$$

Indeed $\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{x}) = L\,\mathbb{E}[\mathbf{z}\mathbf{z}^{\mathsf{T}}]\,L^{\mathsf{T}} = LL^{\mathsf{T}} = \Sigma$, and a linear map of a Gaussian is Gaussian.

> **Modern connection — Gaussian noise drives generative models**
>
> Equation (5) separates the randomness ($\mathbf{z}$) from the parameters ($\boldsymbol{\mu}, L$). Variational autoencoders exploit this "reparameterisation trick" to differentiate through sampling [4]. Diffusion models, behind many of today's image generators, go further. Their *forward process* gradually turns data into noise: with $\bar{\alpha}_t$ decreasing from 1 to 0,
>
> $$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, I),$$
>
> so $\mathbf{x}_t \mid \mathbf{x}_0 \sim \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)I)$, an isotropic Gaussian (Figure 3). A neural network is trained to predict $\mathbf{z}$ from $\mathbf{x}_t$; running the process backwards from pure noise then generates new samples [3, 7]. Everything in this section — isotropic covariances, scaling and shifting, sampling by (5) — is used literally.

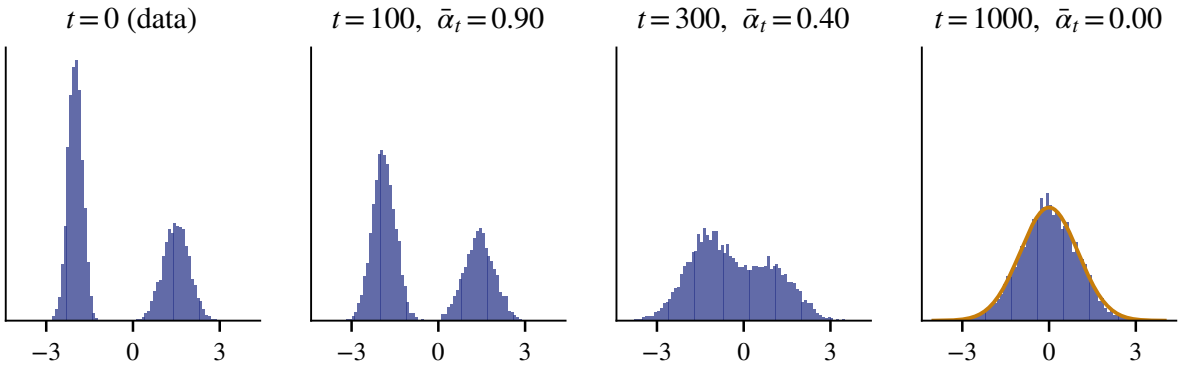


Figure 3. The forward process of a diffusion model applied to a one-dimensional, two-peaked data distribution (histograms of 10^4 samples, the noise schedule of Ho *et al.* [3] with $T = 1000$ steps). By the last step the data are indistinguishable from $\mathcal{N}(0, 1)$ (curve).

6 Learning the parameters

Given $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$, the maximum-likelihood estimates of a Gaussian are the empirical mean and covariance,

$$\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i, \qquad \hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} (\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^{\mathsf{T}} = \frac{1}{n} X_c^{\mathsf{T}} X_c, \tag{6}$$

where X_c is the design matrix of Note 1 with $\hat{\boldsymbol{\mu}}$ subtracted from every row. In one dimension the derivation takes two lines: setting the derivatives of $\sum_i \left[(x_i - \mu)^2/(2\sigma^2) + \tfrac{1}{2}\log\sigma^2\right]$ with respect to μ and σ^2 to zero gives $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$. The variance estimate is slightly biased, $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$, which is why statistics software often divides by $n - 1$; for the sample sizes of machine learning the difference is negligible.

When n is not much larger than d. The matrix $\hat{\Sigma} = X_c^{\mathsf{T}} X_c / n$ has rank at most $n-1$, so for $n \le d$ it is singular: the fitted Gaussian is flat in some directions and (3) cannot be evaluated. Even for n somewhat larger than d its small eigenvalues are badly underestimated. The usual remedies are to restrict the family (diagonal or isotropic) or to *shrink*, $\hat{\Sigma} + \epsilon I$ — the simplest instance of the regularisation discussed in Note 4.

Computing log-densities. In practice one evaluates $\log \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma)$, never the density itself, which underflows in high dimension. With $\Sigma = LL^{\mathsf{T}}$, solve the triangular system $L\mathbf{u} = \mathbf{x} - \boldsymbol{\mu}$; then $\Delta^2 = \|\mathbf{u}\|^2$ and $\log|\Sigma| = 2\sum_i \log L_{ii}$. No matrix is ever inverted.

```
import numpy as np
from scipy.linalg import solve_triangular
def gaussian_logpdf(X, mu, Sigma):            # X: (n, d)
    L = np.linalg.cholesky(Sigma)
    U = solve_triangular(L, (X - mu).T, lower=True)     # (d, n)
    d = X.shape[1]
    log_det_half = np.log(np.diag(L)).sum()             # = 0.5 log|Sigma|
    return -0.5 * (U**2).sum(0) - log_det_half - 0.5 * d * np.log(2 * np.pi)
```

Modern connection — Mahalanobis distance as an anomaly detector

A trained classifier can be fooled by inputs unlike anything it has seen. A simple and strong defence fits a Gaussian to the network's internal feature vectors for each class (with a shared covariance) and flags a new input whose features have a large Mahalanobis distance to every class mean [5]. The same recipe — fit a Gaussian to normal operation, alarm on large Δ^2 — is used to monitor detectors and industrial processes; under the model, Δ^2 follows a χ^2 distribution with d degrees of freedom, which sets the alarm threshold.

Summary A distribution is given by a c.d.f., a probability table or a density; marginals, conditionals and Bayes' rule relate its parts. We sample from, evaluate, infer with and learn distributions, and maximum likelihood is the learning principle — for language models as for χ^2 fits. The multivariate Gaussian is described by a mean and a covariance whose eigenvectors and eigenvalues give the axes and widths of its elliptical contours. It is closed under marginalising and conditioning, sampled as $\boldsymbol{\mu} + L\mathbf{z}$, and fitted by the empirical mean and covariance, which require care when d is comparable to n.

Exercises

E1. Show that the maximum-likelihood estimate of a categorical distribution from counts $n_1, \dots, n_m$ is $\hat{\pi}_k = n_k / n$. (Maximise $\sum_k n_k \log \pi_k$ subject to $\sum_k \pi_k = 1$ with a Lagrange multiplier.) What does it predict for a category never observed, and why is this a problem for a language model?

E2. Verify that (2) integrates to one and has mean μ and variance σ^2.

E3. If the noise has the Laplace density $p(\varepsilon) = \frac{1}{2b} e^{-|\varepsilon|/b}$, which loss does maximum likelihood minimise? Which estimate of a constant θ minimises $\sum_i |y_i - \theta|$, and why is it less sensitive to outliers than the mean?

E4. For $d = 2$, $\boldsymbol{\mu} = 0$ and $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ with $|\rho| < 1$, find the eigenvalues and eigenvectors of Σ and

use (4) to compute $p(x_1 \mid x_2)$. How much does observing x_2 reduce the variance of x_1 when $\rho = 0.9$?

E5. Let $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_d)$. Show that $\mathbb{E}\,\|\mathbf{z}\|^2 = d$ and $\mathrm{Var}\,\|\mathbf{z}\|^2 = 2d$. Conclude that for large d almost all samples lie near a sphere of radius $\sqrt{d}$, far from the mode at the origin. (Note 3 returns to this.)

E6. Implement sampling by (5) and `gaussian_logpdf`. Draw n samples from a Gaussian in $d = 50$ with a random full covariance, refit by (6), and plot the average test log-likelihood against n for the full, diagonal and shrunk ($\hat{\Sigma} + \epsilon I$) estimates. Which wins for small n?

Further reading. Bishop [1], sections 2.1–2.3; Murphy [6], chapters 2–3; for data analysis in physics, Cowan [2].

References

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The selection and order of topics in these notes follow the course IFT 6390 *Fondements de l'apprentissage machine* at Université de Montréal (taught by Ioannis Mitliagkas, slides by Pascal Vincent). The text, examples, figures and exercises of this note are original. Figures are computed from synthetic data by `figures.py`, published with the sources of these notes.